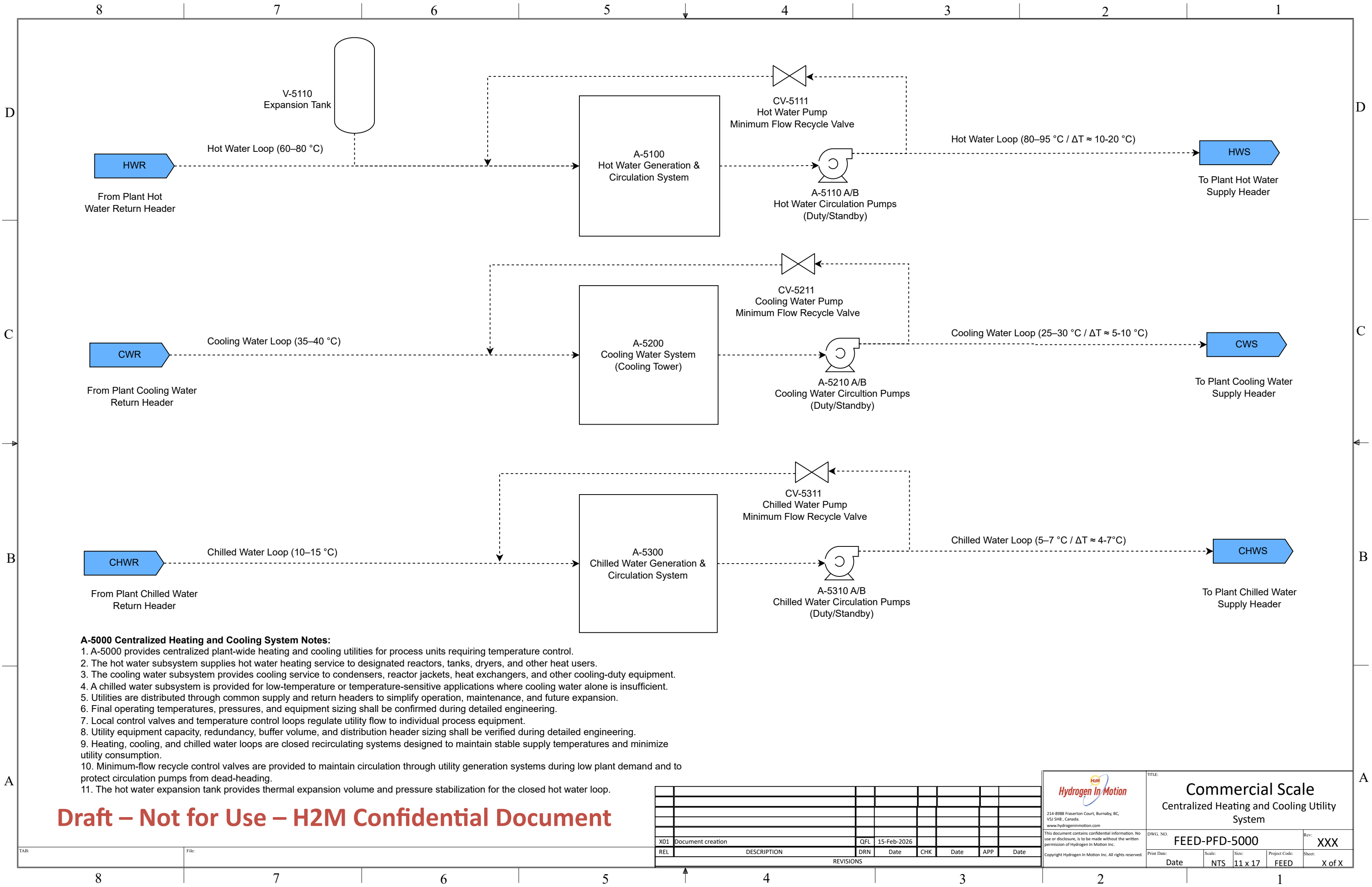




A-5000 Centralized Heating and Cooling System Notes:

- 1. A-5000 provides centralized plant-wide heating and cooling utilities for process units requiring temperature control.
- 2. The hot water subsystem supplies hot water heating service to designated reactors, tanks, dryers, and other heat users.
- 3. The cooling water subsystem provides cooling service to condensers, reactor jackets, heat exchangers, and other cooling-duty equipment.
- 4. A chilled water subsystem is provided for low-temperature or temperature-sensitive applications where cooling water alone is insufficient.
- 5. Utilities are distributed through common supply and return headers to simplify operation, maintenance, and future expansion.
- 6. Final operating temperatures, pressures, and equipment sizing shall be confirmed during detailed engineering.
- 7. Local control valves and temperature control loops regulate utility flow to individual process equipment.
- 8. Utility equipment capacity, redundancy, buffer volume, and distribution header sizing shall be verified during detailed engineering.
- 9. Heating, cooling, and chilled water loops are closed recirculating systems designed to maintain stable supply temperatures and minimize utility consumption.
- 10. Minimum-flow recycle control valves are provided to maintain circulation through utility generation systems during low plant demand and to protect circulation pumps from dead-heading.
- 11. The hot water expansion tank provides thermal expansion volume and pressure stabilization for the closed hot water loop.

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